

Service Report

Jensen, Van Cleve Equipment evaluation

2 – 4 August 2022

Dave Stoller, VP Engineering, Dillon Holtkamp, Engineering

Purpose for visit:

Inspection of PipePro Forms to determine condition and establish cost of refurbishment

Accomplishments in the above listed areas:

- A. Dave Stoller and Dillon Holtkamp were on-site Wednesday, August 4, 2022.
- B. (5) forms were located outside with jackets removed, no lifting means available.
- C. (3) forms were located inside with jackets assembled, no lifting means available.
- D. (5) containers of equipment and joint rings were on the ground with no means of unloading.
 - (1) semi-trailer of equipment and joint rings was present with no means of unloading
- E. Not every item was readily accessible in the containers and trailer.
- F. The pipe forms were to be from the CSI-Wakefield plant.
- G. A set of manhole forms appear to have been from the Forterra Stouffville plate.
- H. We were unable to test any of the vibrators due to lack of power.
- I. We were unable to confirm condition of P-Lock valves and cylinders due to lack of air pressure.
- J. We were unable to check the assembled cores for dents or fatigue cracks.
- K. Turntable anchor pins are missing from all forms.
- L. Core-to-jacket anchor brackets missing on all forms.
- M. Most forms are rather dirty and rusty and need to be thoroughly cleaned before refurbishment.
- N. There was an orphan 30" x 8' PipePro jacket on-site – unidentified origin.

Details listed:

1. 12" Quad

1.1. Core

- 1.1.A. Groove at pallet location needs to be filled or replaced.
- 1.1.B. O-rings need replaced
- 1.1.C. Rubber hoses on vibrator cords are dry rotted.

1.2. Jacket

- 1.2.A. AR top plate needs replaced due to separation.
- 1.2.B. Minor fatigue cracks (already repaired) in jacket bottom rib.
- 1.2.C. Minor fatigue cracks in jacket top tie plates.
- 1.2.D. Header clamp wear areas need replaced.
- 1.2.E. P-lock valve and hoses need to be replaced.
- 1.2.F. Filler section is bent, see picture.

2. 15" Quad – The jacket and core were together and could not be separated during our visit.

2.1. Core

- 2.1.A. Core bottom AR is showing 1/16" wear and should have repairs made.
- 2.1.B. O-rings need replaced
- 2.1.C. (3) vibrator cords were damaged and need to be replaced.

2.2. Base

- 2.2.A. Turntable anchor tubes need to be replaced.

2.3. Jacket

- 2.3.A. Wear ring less than 1/16" wear and is ok.
- 2.3.B. AR top plate is separated and needs repaired.
- 2.3.C. Need to replace bottom jacket AR
- 2.3.D. Some fatigue cracks on tie plates – (2) thru shell plate.
- 2.3.E. Header clamp wear areas need replaced.
- 2.3.F. Several p-lock cylinders and fittings were missing.
- 2.3.G. P-lock valve and hoses need to be replaced.
- 2.3.H. P-lock wear pads need to be replaced.

3. 18" Quad

3.1. Core

- 3.1.A. Groove at pallet location needs to be filled or replaced.
- 3.1.B. O-rings need replaced
- 3.1.C. Rubber hoses on vibrator cords are dry rotted.
- 3.1.D. Core #3 anchor cups and pins need replaced, core is loose.

3.2. Jacket

- 3.2.A. AR top plate needs replaced due to separation.
- 3.2.B. Header clamp wear areas need replaced.
- 3.2.C. Minor fatigue cracks in jacket.
- 3.2.D. Forklift handling dents may need repaired.
- 3.2.E. P-lock valve and hoses need to be replaced.
- 3.2.F. P-lock cylinders seem rusty and should be replaced.

4. 24" Dual

4.1. Core

- 4.1.A. Groove at pallet location needs to be filled or replaced.
- 4.1.B. O-rings need replaced
- 4.1.C. Core to base connections need to be replaced.
- 4.1.D. New isolators, anchor shafts, tubes, pins,
- 4.1.E. (4) vibrator cords were damaged and need to be replaced.

4.2. Base

- 4.2.A. Replace turntable anchor tubes
- 4.2.B. Repair base in area of anchor tubes – ½" plate separated from channels.

4.3. Jacket

- 4.3.A. AR top plate needs replaced due to separation.
- 4.3.B. Header clamp wear areas need replaced.
- 4.3.C. Replace all p-lock wear items.
- 4.3.D. Replace all p-lock cylinders.
- 4.3.E. P-lock valve and hoses need to be replaced.

5. 30" Single

5.1. Core

- 5.1.A. Top AR needs to be replaced.
- 5.1.B. O-ring need replaced

5.1.C. vibrator cord was damaged and need to be replaced.

5.2. Base

5.2.A. Replace turntable anchor tubes

5.2.B. Needs new rubber isolators.

5.2.C. Base pick-ups need replaced

5.2.D. Needs all new core-to-base wear pads and re-mated.

5.3. Jacket

5.3.A. Jacket top is bent in area of front header clamp pin.

5.3.B. Header clamp wear areas need replaced.

5.3.C. Jacket top wear ring needs replaced.

5.3.D. Jacket bottom wear ring needs replaced.

5.3.E. Replace all p-lock wear items.

5.3.F. P-lock valve and hoses need to be replaced.

5.3.G. Jacket rubber seal needs replaced.

6. 36" Single - this form was assembled, and we did not have means to separate it.

6.1. Core

6.1.A. Small shim stock was welded to core top. This needs to be removed but core top is ok.

6.1.B. Base lifting points are worn and need to be replaced.

6.1.C. (2) vibrator cords were damaged and need to be replaced.

6.2. Jacket

6.2.A. Replace (4) p-lock cylinders.

6.2.B. P-lock valve and hoses need to be replaced.

6.2.C. P-lock wear pads need to be replaced.

6.2.D. P-lock air supply damaged and needs replaced.

7. 42" Single - this form was assembled, and we did not have means to separate it.

7.1. Core

7.1.A. Top AR needs to be replaced.

7.1.B. O-ring need replaced

7.1.C. Groove at pallet location needs to be filled or replaced.

7.1.D. Base lifting points are worn and need to be replaced.

7.1.E. (2) vibrator cords were damaged and need to be replaced.

7.1.F. Core has old style anchorage and needs to be remated to base with new isolators.

7.2. Base

7.2.A. Needs (4) rubber isolators replaced.

7.3. Jacket

7.3.A. Jacket rubber seal needs replaced.

7.3.B. P-lock valve and hoses need to be replaced.

7.3.C. AR top plate needs replaced due to separation.

8. 48" Single

8.1. Core

8.1.A. Groove at pallet location needs to be filled or replaced.

8.1.B. O-ring need replaced

8.1.C. (2) vibrator cords were damaged and need to be replaced.

8.1.D. Core has old style anchorage and needs to be remated to base with new isolators.

8.2. Base

- 8.2.A. Replace turntable anchor tubes
- 8.2.B. Needs new rubber isolators.
- 8.2.C. Base pick-ups need replaced

8.3. Jacket

- 8.3.A. AR top plate needs replaced due to separation.
- 8.3.B. Replace all p-lock wear items except cylinders.
- 8.3.C. P-lock valve and hoses need to be replaced.
- 8.3.D. Jacket seal needs to be replaced.

9. Form Summary

- 9.1. 18" 24" and 30" forms have been run pretty hard.
- 9.2. The 24" has some damage to the base.
- 9.3. The 30" needs to be re-mated.
- 9.4. The pallet lock hoses and valves need to be replaced on most all jackets.
- 9.5. We recommend the vibrator cords should be replaced on all forms.

10. Header Clamps

- 10.1. In general, the clamps should have linkages and feet replaced
- 10.2. The clamps should have electrical components replaced
- 10.3. Optionally, all hoses should be replaced.
- 10.4. 12" Quad -
- 10.5. 15" Quad
- 10.6. 18" Quad -
- 10.7. 21" Dual – one clamp foot welded up
- 10.8. 24" Dual
- 10.9. 24" Dual – old style
- 10.10. 30" Single – missing trowel cylinder, clamp cylinders all need replaced. New center bearing
- 10.11. 30" Single – old style
- 10.12. 36" Single – trowel cylinder gone, needs new hydraulic hoses
- 10.13. 42" Single
- 10.14. 48" Single
- 10.15. 30" elliptical dual clamp – old style

11. Platform Filler Sections

- 11.1. 12" quad filler, minor bending
- 11.2. 18" quad filler, minor bending
- 11.3. 21" dual
- 11.4. 24" dual
- 11.5. 42" single
- 11.6. 48" single
- 11.7. We found at least (2) other modified filler section of unknown size and origin

12. Joint Rings

- 12.1. 12" Headers - aluminum
 - 12.1.A. (10) 60301C, 12 1/16" ID, 15.8125" OD
- 12.2. 15" Headers - aluminum

- 12.2.A. (10) 60302C, 15.08 ID
- 12.2.B. 15" Pallets
- 12.2.C. (13)
- 12.3. 15" Pallets**
 - 12.3.A. (60) 28936, 15.08 ID, 24 3/16 OD, 3.5" JT DECENT SHAPE USEABLE
- 12.4. 18" Headers - aluminum**
 - 12.4.A. (10) 60440C, 18.08 ID very good shape.
- 12.5. 18" Pallets**
 - 12.5.A. (105) 58309 – almost new 3.5" – reorder to match original
 - 12.5.B. 24" Header
 - 12.5.C. (2) 20193C worn badly 24.08 ID 29.50 OD 3 1/2" joint **NOT ORIGINAL EQUIPMENT**
 - 12.5.D. 24" Pallets
 - 12.5.E. (16) worn badly 24.25" ID 3 1/2" joint
- 12.6. 24" Pallets**
 - 12.6.A. (19) – 28411C, 3 1/2" JT, original equipment, 24 3/16" ID – pretty worn!!
- 12.7. 24" Pallets**
 - 12.7.A. (16) 28411, 24.125 ID, 33.625 OD 3.5" jt, fairly worn, OD damaged
- 12.8. 30" Headers**
 - 12.8.A. (29) 30.08 ID, 36 17/32 OD, original, worn but useable
- 12.9. 30" Pallets**
 - 12.9.A. (23) 3.5" JT, 30 3/16" ID, fair condition, worn but useable
- 12.10. 30" Pallets**
 - 12.10.A. (12) original equipment, 3 1/2" joint
- 12.11. 36" Headers**
 - 12.11.A. (18) 3 1/2" joint, 36.08 ID, 43.5625 OD
- 12.12. 36" Headers**
 - 12.12.A. (10) 36.08 ID, 43.5" OD, 3 1/2" joint, original, decent condition and useable
- 12.13. 36" Pallets**
 - 12.13.A. (12 +) 28413, 3.5" joint, original
 - 12.13.B. (11) 49236 1" OD – reorder to match original
- 12.14. 42" Headers**
 - 12.14.A. (10) 32374C, 42.125 ID, worn but useable, 4" joint
- 12.15. 42" Pallets**
 - 12.15.A. (10) 30722C, 4" joint, 3/4" flange, 42.125 ID, 53.6875 OD
- 12.16. 48" Headers - pipe**
 - 12.16.A. (11) 38883C 48.125" ID, 4 1/4 JOINT 3/4 bottom flange decent condition
- 12.17. 48" Pallets**
 - 12.17.A. (10) 39344C 48.125" ID 60.875 OD, 4 1/4 JOINT 3/4 bottom flange worn but usable
- 13. Joint Ring Summary**
 - 13.1. 12" needs all new headers and pallets
 - 13.2. 15" needs all new headers – 60 pallets are good and balance needed
 - 13.3. 18" needs all new headers – 105 pallets are good and balance needed
 - 13.4. 24" needs all new headers and we recommend all new pallets

- 13.5. 30" – 29 headers and 35 pallets are worn but serviceable
- 13.6. 36" – 28 headers and 23 pallets are in decent condition
- 13.7. 42" – 10 headers and 10 pallets are worn but serviceable
- 13.8. 48" – 11 headers and 10 pallets are worn but serviceable

14. Manhole Equipment Summary

- 14.1. Joint rings are not directly compatible with manhole riser equipment.
- 14.2. The 48" presshead clamp will not work properly to press headers.
- 14.3. We could not determine if the homemade forms produce a quality product.

15. 48" manhole headers – not directly compatible with manhole risers – Hanson, Sioux Falls

- 15.1. (31) newer with machined tops, bevels, 57468 2 5/8", 48 1/8" ID 57.8 OD

16. 48" manhole pallets – not directly compatible with manhole risers – Hanson, Sioux Falls

- 16.1. (28) 2 3/4 groove, 1" flange with bevel, 57467 fairly new

17. 48" PipePlus Riser – machined jacket tops – from Forterra Stouffville

- 17.1. 4' core and jacket 12380
- 17.2. 3' jacket and stand 12381
- 17.3. 2' jacket and stand 12382
- 17.4. 1' jacket and stand 12383
- 17.5. 48" x 12" flat top 12388 – homemade core and support
- 17.6. 48" Vibrating table for flat tops
- 17.7. 48" x 4' raiser stand (56.25")
- 17.8. 48" x 2' raiser stand (32.25")

18. Homemade forms

- 18.1. 48x3 homemade cone
- 18.2. 48x2 homemade cone
- 18.3. 48x3 homemade riser
- 18.4. Homemade 48" round to 24 square transition form and homemade vibration table
- 18.5. Core roof plates and press plates for IPM monobase – homemade form

19. PipePro Machine was in pieces but mostly accounted for. We did not find the center pipe but it may have been inaccessible in a container. I believe a full controls upgrade would be required

20. 4 kiln posts

21. 4 kiln strongbacks in (2) lengths

22. (2) kiln curtains folded, condition could not be established

23. (?) kiln yoke arms buried in container

24. PipePro Extruder Beams

- 24.1. Beam in container, no manhole chains, new springs, pneumatics lines and equipment and airbags.
- 24.2. Beam outside, no manhole chains, new springs, pneumatics lines and equipment and airbags.

Work outstanding and recommendations:

Discuss with Jensen the details of the report and work through the many variations of what could happen next. It would be preferred to first narrow the scope of the proposed future, to create a budget/estimate/quotation for this equipment.

